OEIS A005315: evidence for M35-M37

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The currently approved public b-file ends at n=34. The three values below are proposed additional terms. M38 is still running and is excluded from this package.

Completed exact computations

Order	Full decimal value	Wall time	Peak sampled RSS	Stages
35	21143326800009460237214623843856 6	31603.365 s	828,465,958,912 B	62
36	23582618387517290192123441049260 12	2191.354 s	945,597,001,728 B	64
37	26370652969449510879209096913242792	5689.257 s	1,771,293,687,808 B	66

The runs used 16 OpenMP threads. Recorded matrix-file reads and writes were zero at every stage. Peak RSS is process memory, distinct from the logical coefficient-array accounting. The three runs used different algorithm revisions and orders; their times are not a controlled speed comparison.

Modular reconstruction for M35 and M36

Each order was computed again in two separate modular runs, modulo 2^64 and modulo (2^64-1). The moduli are coprime; primality is not asserted. Their product, 340282366920938463444927863358058659840, exceeds Catalan(n)^2 for both n=35 and n=36. The uniquely reconstructed integer in that bound matches the exact computation.

Order	Pass 1	Pass 2	Peak RSS: pass 1 / pass 2	Result
35	1764.104 s	1860.789 s	0.656 / 0.655 TB	Exact match
36	3790.724 s	4027.227 s	1.342 / 1.343 TB	Exact match

Scope: the modular runs are fresh numerical computations, but they share the deterministic tree grammar and frontier geometry with the candidate algorithm. They check the arithmetic and CRT reconstruction, not those shared mathematical components independently. No CRT run has been completed for M37.

Method, provenance and submission limits

The implementation represents lower noncrossing matchings by a deterministic weighted tree grammar and propagates exact integer coefficients through frontier geometry. Later revisions construct geometry with bitmaps and keep sparse supports. Checked per-block capacity bounds are applied before numerical writes. Matrices remained in anonymous RAM; no matrix spill files were used.

Validation records

- M35: complete exact run with 62 recorded stages and two modular runs with matching CRT reconstruction.

- M36: complete exact run with 64 recorded stages and two modular runs with matching CRT reconstruction.

- M37: complete exact run with 66 recorded stages. The source, phase sums, arithmetic caps, sampled RSS and zero matrix I/O were audited. This term has no CRT check.

- Small-order tests, coefficient comparisons and the raw JSON/stage records are retained in the accompanying source release and evidence packet. File hashes attest byte integrity, not institutional certification of remote execution.

Frozen source identifiers

M37 source manifest SHA-256: f6835ac33bde4f40d28355e0b81e2ac2c7368da4fe1bcd491a1461e6d3b28dd5

M35-M36 modular source manifest SHA-256: e55b0bde972c0c53dd8f4c363416ec91e66958384f0f5f56ad67c0f7d5b1cc7a

The separate tree-code publication candidate has passed local package checks but has not yet been published to a new GitHub repository or Zenodo record. No new public URL or DOI is claimed here. This packet supplements its earlier validation notes with the subsequently completed M35-M36 modular runs.

For the OEIS author

The accompanying b005315.txt preserves the approved n=0..34 prefix exactly and adds only n=35..37. The human author must personally review the values, method and evidence, and write any OEIS editorial comments in their own words. Retain the existing attribution to Vladimir Osipov for the earlier M29-M30 reports. Upload the replacement b-file and add the link generated by the OEIS form.

OEIS A005315: https://oeis.org/A005315 | b-file instructions: https://oeis.org/SubmitB.html | AI policy: https://oeis.org/wiki/Use_of_AI_for_OEIS_Submissions

Computational evidence is not a formal proof of every program path. M37 remains a candidate extension with one complete exact run and internal checks. M38 is excluded until its calculation and audit finish.